

# Kitchcu

## CPO Product Blueprint

Modules | Functionalities | Data Flows | Application Flows | Pain Points | Solutions | Roadmap

Kitchcu cloud kitchen platform

**India's first - and the world's third - platform with this feature stack**

*Version 4.2 | July 2026 | Confidential*

## Product North Star

Kitchcu is the operating system for cloud kitchens - not an aggregator. Subscription SaaS with zero food commission. Owners run everything from one PWA; customers trust what they order through live media and home-taste ratings. Positioning: India's first - and the world's third - platform with this feature stack.

**WhatsApp to revenue**

Owner promise

**Live photos + fair delivery**

Customer promise

**Subscription, 0% food**

Platform model

**10 pilot kitchens / 3 mo**

MVP target

### Non-Negotiable Principles

- Truth in media - hero dish photos must be live-capture (no stock deception)
- Quality over speed - owner-set prep/delivery windows, never fake 10-min races
- Owner sovereignty - CRM, coupons, and customer history belong to the kitchen
- Progressive complexity - advanced features unlock after kitchen traction (50+ orders)
- WhatsApp-native - meet owners where 70%+ of orders already happen

# Owner Pain Points (Market Reality)

| Pain                              | Business Impact                       | Severity |
|-----------------------------------|---------------------------------------|----------|
| Orders trapped in WhatsApp chats  | Lost orders, zero analytics           | Critical |
| Aggregator 18-30% commission      | Margin erosion, no customer ownership | Critical |
| No daily profit visibility        | Cash flow guesswork, bad decisions    | High     |
| Stock photos mislead customers    | Refunds, reputation damage            | High     |
| Inconsistent taste batch-to-batch | Repeat customer loss                  | High     |
| No CRM or targeted marketing      | Cannot win back regulars              | Medium   |
| Promotions based on guesswork     | Wasted discounts                      | Medium   |
| Walk-in + call + chat silos       | Double booking, chaos                 | Medium   |

# Customer Pain Points

| Pain                     | Why It Matters             | Kitchcu Answer                 |
|--------------------------|----------------------------|--------------------------------|
| Cannot trust menu photos | Expectation vs reality gap | Live-capture dish media        |
| Opaque delivery fees     | Cart abandonment           | PostGIS distance + owner rules |
| Generic star ratings     | No home-food signal        | Home-taste benchmark 1-5       |
| One kitchen per cart     | Inconvenient lunch combos  | Multi-kitchen checkout (P2)    |
| No order transparency    | Anxiety, support calls     | Lifecycle + WhatsApp updates   |
| No tiffin subscription   | Daily meal friction        | Kitchen meal plans (P2)        |

"I run 80 orders a day on WhatsApp and still do not know my profit." - Typical owner

# Pain Point to Solution Mapping

| Pain ID | Problem               | Kitchcu Module       | Solution                                      |
|---------|-----------------------|----------------------|-----------------------------------------------|
| P1      | WhatsApp chaos        | Order + Notification | Unified inbox, parser, manual fallback        |
| P2      | Aggregator commission | Billing              | Flat subscription; kitchen keeps food revenue |
| P3      | No profit view        | Analytics            | Revenue, dish, pattern reports                |
| P4      | False menu photos     | Catalog + Media      | Live-capture enforced for hero images         |
| P5      | Taste inconsistency   | Catalog + Ingredient | Per-dish standards, stock mapper (P3)         |
| P6      | No customer ownership | Marketing            | Owner CRM, coupons, tiffin                    |
| P7      | Guesswork promos      | Growth Engine        | Seasonal, win-back, combo suggestions         |
| P8      | Multi-channel chaos   | Order Service        | Single lifecycle for all sources              |

# Platform Architecture (Microservices)



Event-driven design: every write publishes EventEnvelope after DB commit.

# Platform Modules & Build Status

| Module            | Responsibility                              | Schema         | Status     |
|-------------------|---------------------------------------------|----------------|------------|
| Gateway           | Auth routing, service proxy                 | -              | DONE v0.3  |
| Identity          | Owners, kitchens, OTP, JWT                  | ckac_identity  | DONE S1    |
| Catalog           | Dishes, categories, live media              | ckac_catalog   | DONE S2    |
| Order             | Manual orders, lifecycle, history           | ckac_orders    | DONE S3    |
| Notification      | WhatsApp, push, SMS, support chat & tickets | ckac_support   | S4 LIVE    |
| Billing           | Subscriptions, UPI, split pay               | ckac_billing   | Sprint 6   |
| Analytics         | Owner revenue, segments, peak hours         | order svc      | S5 PARTIAL |
| Marketing         | CRM, coupons, tiffin                        | ckac_marketing | Phase 2    |
| Rating            | Home taste, A/V reviews                     | ckac_ratings   | Phase 2    |
| Delivery          | Radius, fees, tracking                      | geo rules      | Phase 2    |
| Growth            | AI-ready suggestions                        | ckac_growth    | Phase 2-3  |
| Learning / Stream | Recipes, live prep                          | ckac_learning  | Phase 3    |

# Module: Identity Service

## Functionalities

- Owner registration (phone + OTP)
- JWT access + refresh tokens
- Kitchen onboarding with PostGIS location
- Kitchen code generation (e.g. CKPNQ001)
- Owner-kitchen authorization checks

## Data Model (ckac\_identity)

- owners: id, phone, name, subscription\_tier, status
- kitchens: id, owner\_id, code, name, location (geography)

## API Endpoints (via Gateway)

- POST /api/v1/owners/register
- POST /api/v1/auth/otp/request | verify
- POST /api/v1/kitchens
- GET /api/v1/kitchens/me



# Module: Catalog Service

## Functionalities (F13-F15)

- 10 default categories seeded per kitchen
- Dish CRUD with price, prep time, quality notes
- Live-capture rule: hero image must be is\_live\_capture
- Public menu endpoint (cacheable)
- Publishes dish.created / dish.updated events

## Events

dish.created -> ckac:catalog:dish stream

## Business Rules

- No stock photo as hero - CPO trust mandate
- Cross-schema read: verify kitchen ownership
- No cross-schema writes
- Menu invalidation on dish.updated (Redis)

# Module: Order Service

## Functionalities (F03-F05, F30)

- Manual order creation (walk-in, phone)
- Line items from catalog (price snapshot)
- Lifecycle state machine with audit trail
- Order history with status/source filters
- Per-dish prep time drives ETA

## Order Code Format

- {kitchen\_code}-BILL-{YYYYMMDD}-{SEQ}
- Example: CKPNQ001-BILL-20260712-0001
- Bill ID: BILL-20260712-0001

## Lifecycle States

received -> accepted -> preparing -> ready -> out\_for\_delivery -> delivered  
Any pre-delivered state -> cancelled (reason required)  
Each transition -> order\_status\_events row + order.status.changed event

# Planned Modules (Sprints 4-6 & Phase 2)

| Module          | Key Features                           | Events / Integrations             |
|-----------------|----------------------------------------|-----------------------------------|
| Notification S4 | WhatsApp webhook, message parser, push | Consumes order.*; F01-F02, F45    |
| Billing S6      | Owner subscription, UPI, COD, Razorpay | payment.captured; F26, F42-F44    |
| Analytics S6+   | Revenue, best dishes, patterns         | Consumes order.delivered; F07-F12 |
| Marketing P2    | CRM, coupons, tiffin, daily push       | F34-F40                           |
| Delivery P2     | Radius, fee quote, tracking links      | PostGIS; F27-F31                  |
| Rating P2       | Home taste + A/V anonymous reviews     | F16-F18                           |
| Growth P2       | Seasonal, win-back, combo suggestions  | F11                               |

## Owner Application Flow - Day 1 to Daily Ops

### DAY 1 ONBOARDING

Register phone -> OTP (JWT) -> Create kitchen (geo pin)  
-> Add 5 dishes (live camera) -> Publish menu  
-> Connect WhatsApp Business (S4) -> First order in inbox

### DAILY OPERATIONS

Home Inbox: new orders + WhatsApp drafts + today stats  
-> Tap order -> Confirm / edit items -> Advance lifecycle (1 tap)  
-> Customer notified (WhatsApp + push)  
-> End of day: revenue report, top dish, growth tip

### OWNER PWA NAV (target)

Home | Orders (Active/History/Drafts) | Menu | Customers | Analytics  
| Marketing | Subscriptions | Payments | Settings

## Customer Application Flow (Phase 2 PWA)

### DISCOVERY & ORDER

Open PWA link or Discover map (PostGIS nearby)

- > Filter: distance, rating, live-prep premium
- > Kitchen profile: live photos, home-taste scores
- > Add items (multi-kitchen cart supported)
- > Delivery fee breakdown shown -> Pay UPI or COD
- > Track: received ... delivered (WhatsApp + push)

### POST-ORDER

- > Rate home taste (1-5) + quality + optional 15s video
- > Repeat order in 2 taps from history
- > Subscribe to tiffin plan (optional)

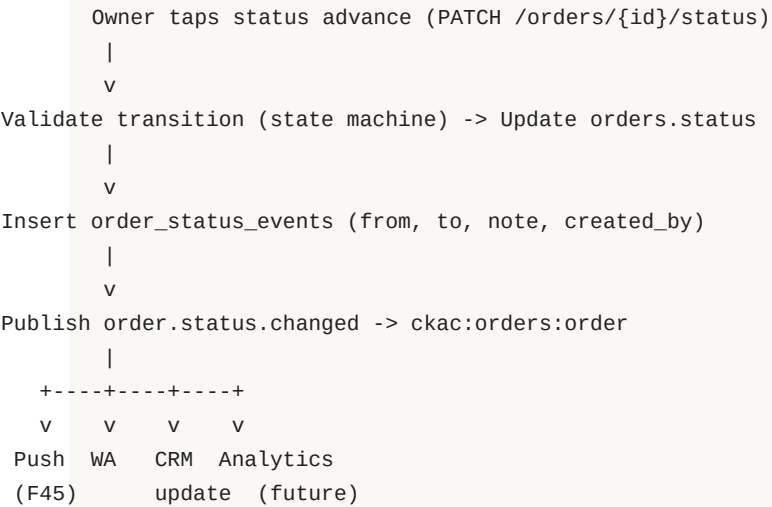
### CUSTOMER PWA NAV

Discover | Cart | Orders & Tracking | Subscriptions | Ratings | Profile

# Order Intake - Multi-Source Data Flow

| SOURCES                                                   |    | INTAKE PATH                      |    | OUTPUT         |
|-----------------------------------------------------------|----|----------------------------------|----|----------------|
| -----                                                     |    | -----                            |    | -----          |
| WhatsApp message                                          | -> | Webhook -> Parser -> Draft order | -> | Owner confirms |
| Phone / walk-in                                           | -> | Owner manual entry (S3 DONE)     | -> | order.placed   |
| Customer PWA                                              | -> | Cart checkout (Phase 2)          | -> | order.placed   |
| order.placed event:                                       |    |                                  |    |                |
| DB commit (ckac_orders) -> Redis Stream ckac:orders:order |    |                                  |    |                |
| -> Notification service (owner alert + customer confirm)  |    |                                  |    |                |
| -> Analytics worker (counters, daily stats)               |    |                                  |    |                |
| -> Billing worker (payment capture when online)           |    |                                  |    |                |

# Order Lifecycle - Event-Driven Data Flow



Cancel requires reason. Terminal states: delivered, cancelled.

# Event Catalog (Event-Driven Design)

| Event                     | Producer     | Stream                | Consumers                  |
|---------------------------|--------------|-----------------------|----------------------------|
| kitchen.created           | identity     | ckac:identity:kitchen | analytics                  |
| dish.created              | catalog      | ckac:catalog:dish     | search, cache              |
| dish.updated              | catalog      | ckac:catalog:dish     | menu cache invalidation    |
| order.placed              | order        | ckac:orders:order     | notify, analytics, billing |
| order.status.changed      | order        | ckac:orders:order     | notify, tracking UI        |
| payment.captured          | billing      | ckac:billing:payment  | settlement, reports        |
| whatsapp.message.received | notification | ckac:notify:wa        | order parser               |

Contract: DB commit first, then publish. Outbox table in ckac\_events for reliability.



## Database Architecture - Schema-per-Domain

| Schema         | Tables                                   | Phase | Status     |
|----------------|------------------------------------------|-------|------------|
| ckac_identity  | owners, kitchens                         | 1     | LIVE       |
| ckac_catalog   | categories, dishes, dish_media           | 1     | LIVE       |
| ckac_orders    | orders, order_items, order_status_events | 1     | LIVE       |
| ckac_events    | outbox, processed_events                 | 1     | LIVE       |
| ckac_billing   | payments, subscriptions, settlements     | 1-2   | Planned S6 |
| ckac_marketing | kitchen_customers, coupons, plans        | 2     | Planned    |
| ckac_ratings   | dish_ratings, suggestions                | 2     | Planned    |
| ckac_growth    | suggestions, seasonal_patterns           | 2-3   | Planned    |

Scaling: 0-500 kitchens single PG; read replica + PgBouncer at 500-5K; partition events by month at 5K+.

# API Gateway Routing & Service Ports

| Path Pattern                                 | Target Service | Port (host) |
|----------------------------------------------|----------------|-------------|
| /api/v1/auth/*, /api/v1/owners/*             | Identity       | 18001       |
| /api/v1/kitchens/{id}/categories menu dishes | Catalog        | 18002       |
| /api/v1/kitchens/{id}/orders/*               | Order          | 18003       |
| /api/v1/orders/*                             | Order          | 18003       |
| /api/v1/kitchens (create/list)               | Identity       | 18001       |
| Gateway entry                                | Gateway        | 18000       |

## Phase 1 Features - Owner Can Run Kitchen

| ID      | Feature                                   | Module        | Sprint | Status  |
|---------|-------------------------------------------|---------------|--------|---------|
| F01     | WhatsApp order capture                    | notification  | S4     | PARTIAL |
| F02     | Message parser                            | notification  | S4     | DONE    |
| F03     | Manual order input                        | order         | S3     | DONE    |
| F04     | Order lifecycle                           | order         | S3     | DONE    |
| F05     | Order history                             | order         | S3     | DONE    |
| F07     | Revenue report                            | analytics     | S6     | Planned |
| F13-F15 | Menu + live photo + categories            | catalog       | S2     | DONE    |
| F26     | Owner subscription                        | billing       | S6     | Planned |
| F30     | Per-dish prep time                        | order+catalog | S3     | DONE    |
| F42-F43 | UPI / COD payments                        | billing       | S6     | Planned |
| F45     | WhatsApp + app notifications + AI support | notification  | S4     | PARTIAL |

## Phase 2 Features - Growth (Months 4-6)

### Customer Experience

- F06 Multi-kitchen single checkout
- F32 Kitchen discovery (map + distance)
- F33 Order history + repeat order
- F16-F18 Home taste ratings + A/V
- F41 Custom cooking requests

### Owner Growth

- F34-F35 Tiffin / meal plans
- F36-F38 CRM, coupons, targeted pricing
- F39 Daily menu WhatsApp push
- F07-F11 Analytics + growth suggestions
- F44 Aggregated payment + split settlement
- F27-F31 Delivery radius, fees, tracking

## Phase 3 Features - Differentiation (Moat)

- F19 Ingredient balance mapper - stock deduct, low-stock alerts
- F20 Customer recipe suggestions (owner accept/reject workflow)
- F21-F23 Learning portal, trial batches, recipe rewards
- F24 Best cloud chef rankings (city / state / national)
- F46-F48 Live kitchen streaming (optional owner opt-in premium)
- F12 Performance report with recipe improvement suggestions

### Ranking Formula (Benchmark)

Monthly score = 30% avg rating + 20% recipe shares + 25% review volume + 15% repeat rate + 10% consistency index.

# Current Build Status - July 2026

S1-S4 complete; S5 partial

Sprints done

5 (+ gateway)

Services live

90+ passing

Automated tests

Portal + 3 PWAs

Apps live

| Sprint | Deliverable                                    | Status       |
|--------|------------------------------------------------|--------------|
| S1     | Identity, gateway, Docker, JWT, TDD foundation | COMPLETE     |
| S2     | Catalog, live-capture, dish.created events     | COMPLETE     |
| S3     | Order manual, lifecycle, order.placed events   | COMPLETE     |
| S4     | WhatsApp webhook + parser + AI support chat    | COMPLETE     |
| S5     | PWAs + portal + analytics + ticketing          | PARTIAL LIVE |
| S6     | Billing + revenue reports                      | Planned      |

## Multi-Kitchen Checkout Flow (F06 - Phase 2)

```
Customer cart:
[Kitchen A: 2 dishes] + [Kitchen B: 1 dish]
  |
  v
master_orders (1 customer payment)
  |
+-----+-----+
v               v
order (CKA-...) order (CKB-...)
separate lifecycle, separate bills, separate tracking
  |
  v
Razorpay Route: automatic split to each kitchen linked account
Customer: unified receipt + per-kitchen tracking tabs
```

## Delivery Fee Decision Flow (F27-F28)

1. Compute distance (PostGIS) owner kitchen <-> customer address
2. If distance <= free\_delivery\_radius -> fee = 0
3. If > radius -> apply owner rules (flat / per-km / min order)
4. Show breakdown at checkout BEFORE payment
5. Customer denies fee:
  - a. Owner can cancel order
  - b. Owner can waive (min order rule)
  - c. Customer switches to self-pickup



# Owner Growth Intelligence Engine (F11)

| Suggestion Type | Trigger                      | Example                                  |
|-----------------|------------------------------|------------------------------------------|
| Seasonal        | Calendar + regional patterns | Diwali in 3 weeks - add mithai combo     |
| Dish promo      | High rating, low volume      | Paneer Tikka 4.8 but 5% orders - promote |
| Win-back        | Regular inactive 21+ days    | 12 customers - send coupon MAAS20        |
| Combo           | Order association mining     | Naan + Dal ordered together 68% - bundle |
| Peak staffing   | Historical volume by hour    | Fri 12-2 PM is 3x avg - prep extra       |

# Rating & Trust Layer (F16-F18)

| Dimension    | Scale     | Purpose                            |
|--------------|-----------|------------------------------------|
| Home taste   | 1-5       | Tastes like authentic home cooking |
| Quality      | 1-5       | Freshness, portion, packaging      |
| Optional A/V | 15-30 sec | Anonymous video/audio review       |

- Ratings only from verified completed orders - no fake reviews
- Aggregated on dish page without customer identity
- Feeds chef rankings and growth suggestions

## Business Model - Subscription, Not Commission

| Tier       | Price/mo | Includes                                   |
|------------|----------|--------------------------------------------|
| Starter    | Rs 499   | WhatsApp orders, menu, basic reports       |
| Pro        | Rs 1,499 | CRM, marketing, tiffin, coupons, analytics |
| Enterprise | Rs 3,999 | Live stream, API, multi-branch             |

- Zero per-order food commission - kitchen keeps 100% food revenue
- Customer tiffin: 0% Kitchcu cut
- Payments: pass-through via Razorpay Route (multi-kitchen split)
- Yearly plans: 2 months free

# CPO Success Metrics & KPIs

| Metric                   | Phase 1 Target       | How Measured         |
|--------------------------|----------------------|----------------------|
| Time to first order      | < 5 min post-signup  | Onboarding funnel    |
| WhatsApp parse rate      | 95% parsed or manual | Parser accuracy log  |
| Lifecycle notify latency | < 2s event to push   | Event SLA monitoring |
| Hero live-capture rate   | 100%                 | Catalog audit        |
| Owner retention M6       | 80%                  | Subscription churn   |
| Repeat customer rate     | 40%+                 | CRM aggregates       |
| Owner GMV on platform    | North star           | Sum of order totals  |

## Competitive Position

- vs Aggregators (Swiggy/Zomato): zero commission, owner owns CRM, live-capture integrity
- vs POS (PetPooja): customer-facing discovery, quality trust layer, WhatsApp analytics
- Unique combination: WhatsApp-native ops + live menu + multi-kitchen cart + home-taste ratings
- Kitchcu occupies high owner-control quadrant WITH customer discovery - not either/or

**Rs 2.5L Cr+**

TAM India delivery GMV

**~50,000**

SAM semi-formal kitchens

**500 kitchens**

SOM Year 3 target

**Rs 9 Cr**

ARR at Rs 1,499/mo

## Go-To-Market Strategy

- Pilot clusters: 10 kitchens in one locality - network effect for customer discovery
- WhatsApp-first onboarding in Hindi and regional languages
- Kitchen-branded PWA links shared on Instagram / WhatsApp status
- Monthly city chef rankings for organic PR and community buzz
- Aggregator escape narrative: Keep 100% of your food revenue
- Progressive feature unlock - simple Day 1, powerful Month 3

# Technology Stack (Scalable Foundation)

| Layer          | Choice                        | Rationale                             |
|----------------|-------------------------------|---------------------------------------|
| Frontend       | PWA React + Vite + TS         | WhatsApp deep links, one codebase     |
| Backend        | Python FastAPI microservices  | Async, OpenAPI, event-driven          |
| Database       | PostgreSQL 16 + PostGIS       | ACID, geo queries, schema isolation   |
| Cache / Events | Redis Streams                 | Menu cache, event bus (Kafka Phase 4) |
| Media          | MinIO / S3                    | Live-capture dish photos              |
| Payments       | Razorpay Route                | Multi-kitchen split settlement        |
| Notify         | WhatsApp Cloud API + Web Push | Dual channel reach                    |
| Infra          | Docker -> Kubernetes          | 500 kitchens today, 50K path          |

## Every cloud kitchen deserves to operate like a brand

- Data instead of guesswork - revenue, patterns, growth suggestions
- Trust instead of stock photos - live capture + home-taste ratings
- Direct relationships instead of rented customers - owner CRM
- Community instead of isolation - chef rankings, recipe rewards

**Kitchcu is infrastructure - not the next food aggregator.**



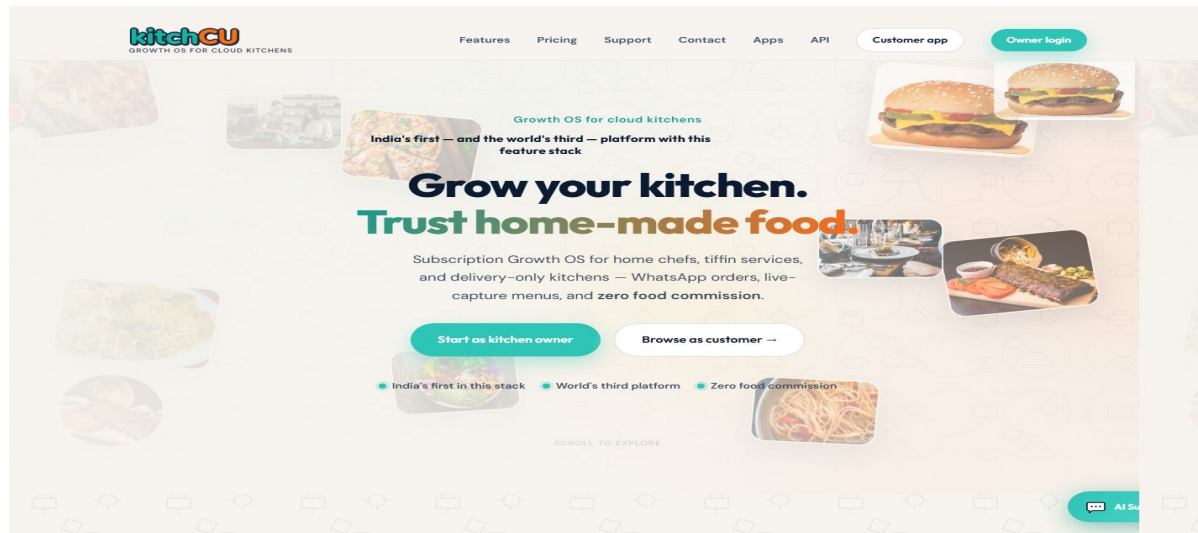
# Shipped Module Status (July 2026)

| Module                     | Challenge solved                     | Status       |
|----------------------------|--------------------------------------|--------------|
| Identity + Catalog + Order | Onboard, live menu, lifecycle        | DONE S1-S3   |
| Notification + Support     | WhatsApp + tickets + tracking nudges | DONE S4/S14  |
| Billing + GST finance      | Subscription + monthly tax audit     | DONE S6+GST  |
| Multi-kitchen + Route pay  | One cart, fair kitchen payouts       | DONE S8-S9   |
| Marketing CRM / coupons    | Owner owns customers                 | DONE S10     |
| Ratings + Growth           | Home taste + actionable suggestions  | DONE S11-S12 |
| Delivery + Ingredients     | Fair fees + pantry deduct            | DONE S13/S15 |
| Learning/Community/Live    | Skills, rankings, opt-in stream      | DONE S16-S18 |
| Quality loop E1+E2         | Purchases + lock chef standards      | DESIGN       |

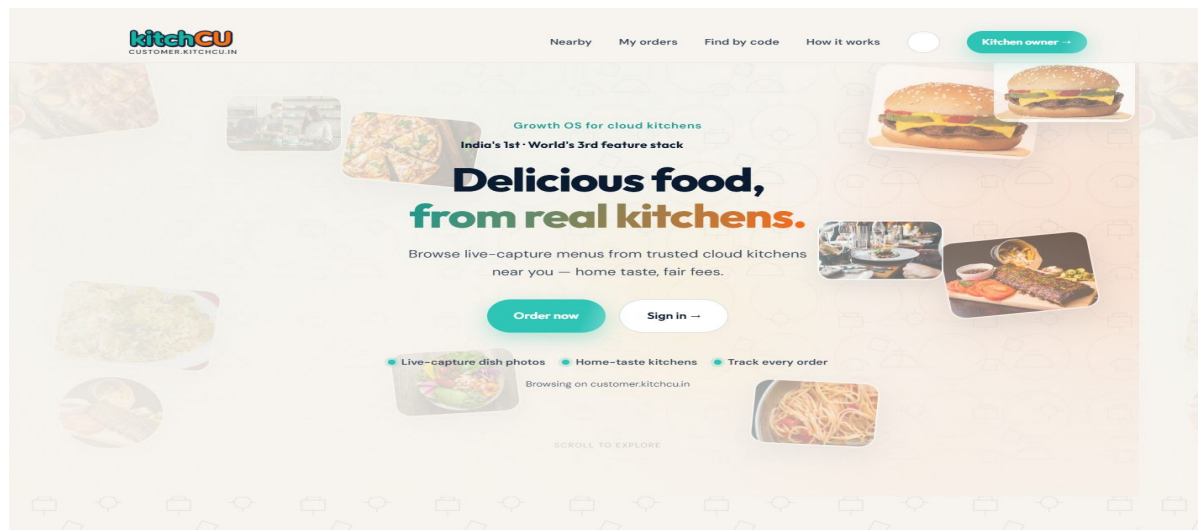
## UI Catalog - Portal & Customer (Light Theme)

Marketing and discovery surfaces use cream/teal/orange. Brand first, one composition, appetite-forward - not ops dashboards.

**Portal home (kitchcu.in) - brand hero + positioning claim + pricing entry**

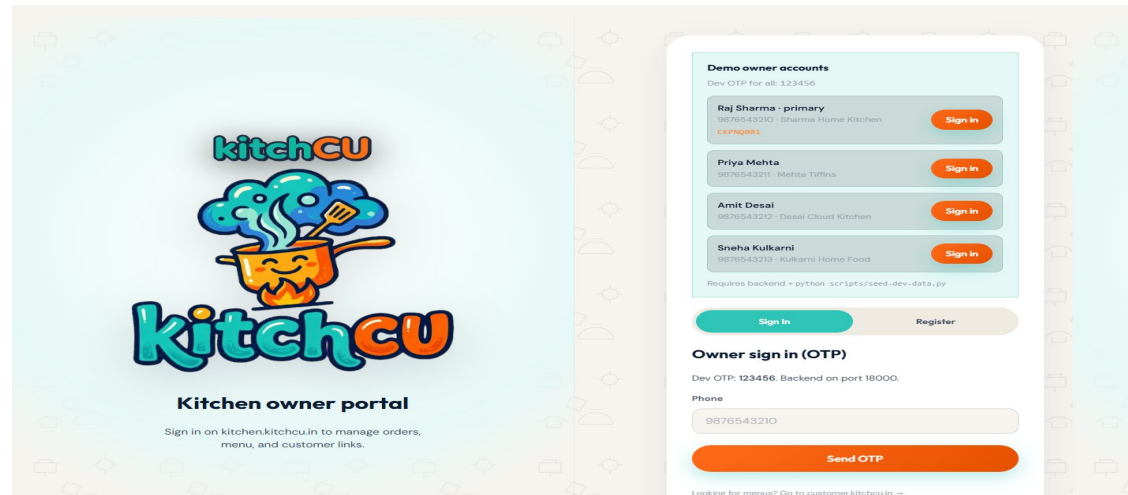


**Customer home (customer.kitchcu.in) - nearby kitchens, live-capture cards, filters**

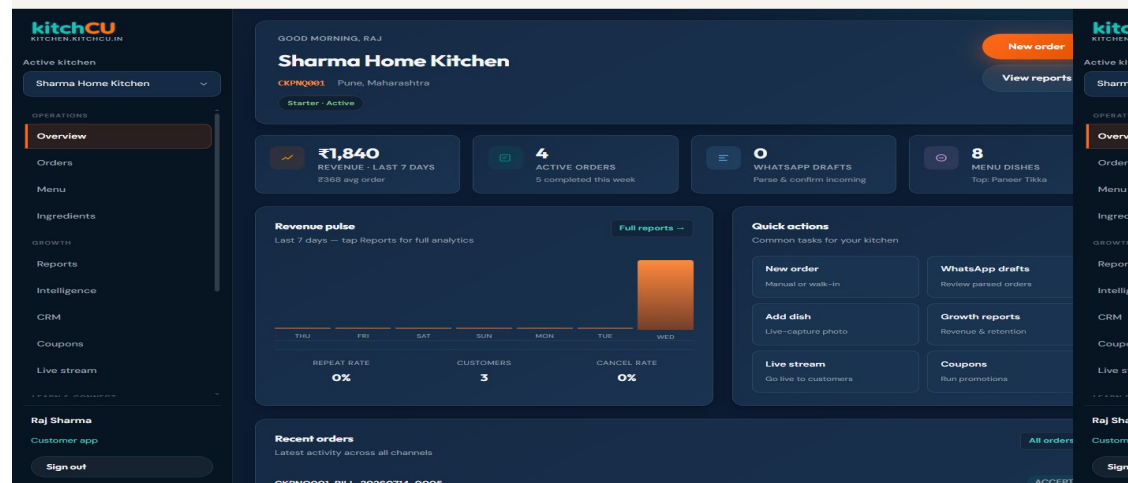


# UI Catalog - Kitchen Login & Owner Ops (Dark Theme)

Kitchen login - OTP phone auth; light cream sheet into dark ops after login



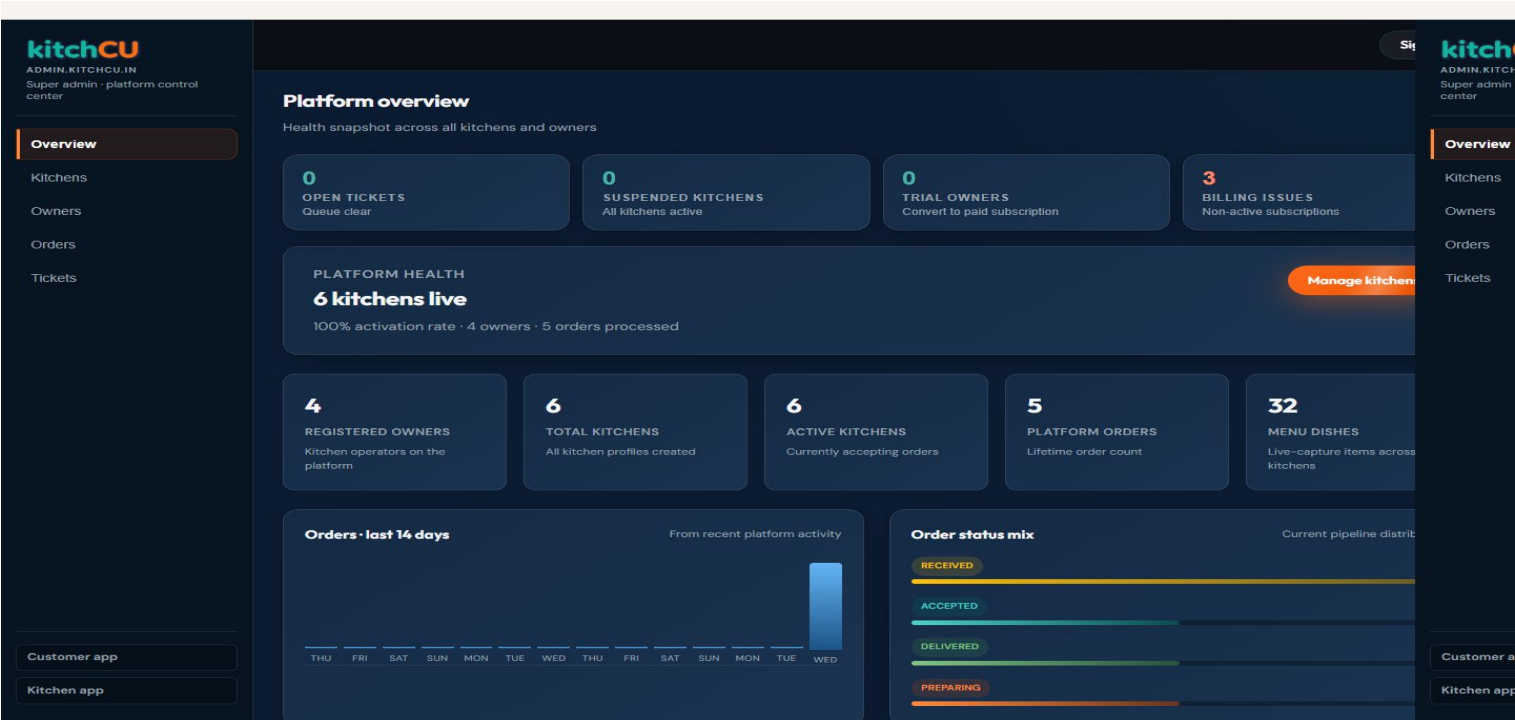
Owner dashboard - dark ops command center; inbox-first, today's revenue + orders



# UI Catalog - Admin Overview

Admin is platform-scope only (never owner-scope mutations). Shares dark ops visual language with the kitchen PWA.

Admin overview (admin.kitchcu.in) - kitchens, orders, tickets at platform scope



## CTO Architecture Snapshot

- Edge: API Gateway routes /api/v1 to 13 domain services (ports 18001-18012)
- Data: PostgreSQL schema-per-domain + PostGIS; Redis Streams + transactional outbox
- Writes never cross schemas; Growth/Ratings read orders cross-schema for evidence only
- PWAs: portal, customer.kitchcu.in, kitchen.kitchcu.in, admin.kitchcu.in
- Full diagrams (architecture, flows, ER, UI anatomy): CKAC-COMplete-GUIDE.pdf v3.1

```
PWAs -> Gateway -> Identity|Catalog|Order|Billing|Marketing|Ratings|
      Growth|Delivery|Learning|Community|Streaming|Notify
      -> PostgreSQL schemas + Redis Streams + MinIO
```

## API & OpenAPI - One Live Contract, Not a Slide Deck

The gateway aggregates every domain service's own FastAPI OpenAPI spec into one contract at request time - the public API docs can never drift from shipped code.

| Surface                | URL                           | Purpose                                   |
|------------------------|-------------------------------|-------------------------------------------|
| Aggregated spec (JSON) | gateway /openapi.json         | Merged, service-prefixed schema           |
| Swagger UI             | gateway /docs                 | Interactive explorer over aggregated spec |
| ReDoc                  | gateway /redoc                | Read-only reference rendering             |
| Portal explorer        | portal /openapi (+ /api-docs) | Same schema inside the marketing shell    |
| Human index            | docs/API.md                   | Auth cheat-sheet + quick-start examples   |

- Every route ships explicit summary/description/responses + Field descriptions - not auto-only
- ?refresh=true on /openapi.json force-refreshes the aggregate after a route change
- Full journey pack for every persona/screen/API call: CKAC-USERFLOWS.md / .pdf

# Kitchcu

Kitchcu cloud kitchen platform

**India's first - and the world's third - platform with this feature stack**

hello@kitchcu.in | kitchcu.in | Pilot waitlist open

*Full guide: CKAC-COMplete-GUIDE.md / .pdf (v3.1) | CPO blueprint v4.2 | User journeys: CKAC-USERFLOWS.md / .pdf | API: docs/API.md + gateway /docs (aggregated OpenAPI) | UI shots: docs/assets/ui/ | E1-E2 quality-loop design pack*